

SVBRDF estimation using a normal sorting technique

Snehal Padhye
sp1471@rit.edu
Rochester Institute of Technology
Rochester, NY, USA

David Messinger
David.Messinger@rit.edu
Rochester Institute of Technology
Rochester, NY, USA

James A. Ferwerda
jaf@cis.rit.edu
Rochester Institute of Technology
Rochester, NY, USA

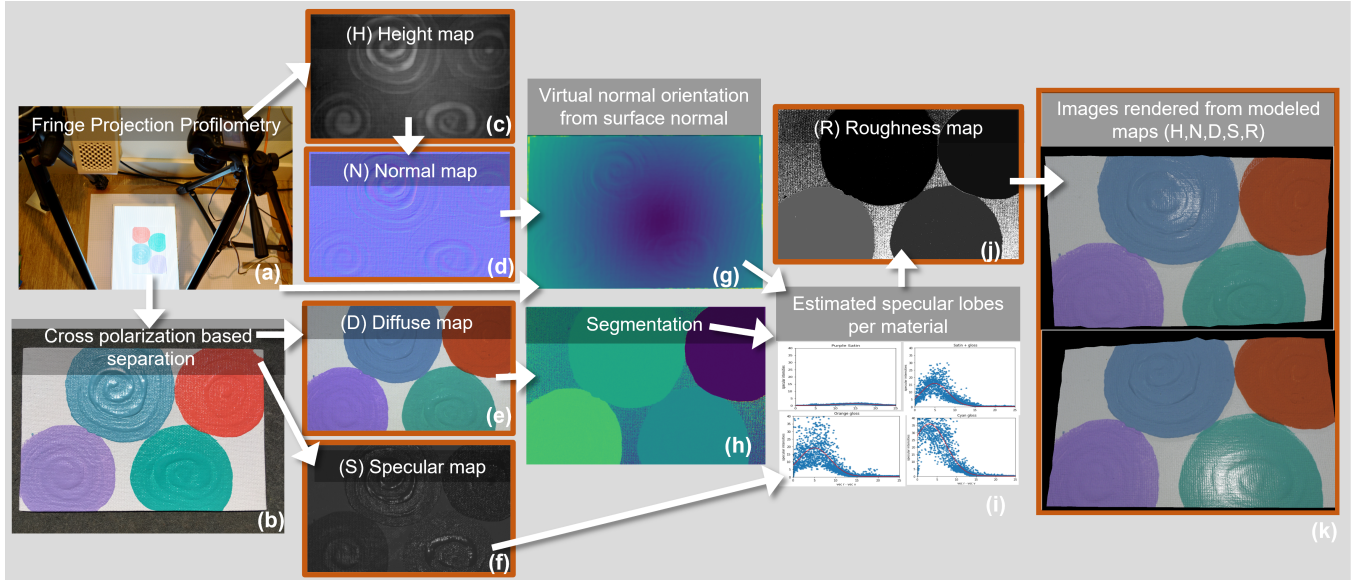


Figure 1: (a) Camera and projector system used to capture sequences of HDR images (b) used to estimate surface characteristics. At this stage fringe projection profilometry is used to estimate surface height and normal maps (c,d). and cross polarization is used to separate the reflection components into diffuse and specular maps (e,f). Next, the geometry of the imaging system (from calibration), and the normal map are used to estimate virtual normal orientations (g), and the diffuse map is used to segment the painting into different materials (h). Specular intensities for each of the materials are then sorted with respect to the virtual normals and fitted with Gaussians (i), to estimate the statistics of their specular lobes, which are then used to create a per-material roughness map (j). The final column (k) shows images of the painting rendered using the estimated height (H), normal (N), diffuse (D), specular (S), and roughness (R) maps.

CCS CONCEPTS

• Computing methodologies → Rendering: Modeling and simulation.

KEYWORDS

Appearance capture, digital modeling, Rendering

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1 INTRODUCTION

Spatially varying bi-directional reflectance distribution functions (SVBRDFs) play an important role in appearance modeling of real-world objects. Automatic capture of these features is highly desirable, and there has been a lot of research on creating practical and lightweight setups for SVBRDF acquisition [Francken et al. 2009][Ferwerda 2018]. Structured light techniques have been widely used for 3D modeling of objects [Geng 2011]. Capturing SVBRDFs using structured light is challenging, which is often the main limitation in using it for modeling the complete appearance of objects. Baek et al. [Baek et al. 2018] used polarimetric images with the structured light setup to measure surface roughness and Rushmeier et al. [Rushmeier et al. 2015] used multiple pairs of cameras and projectors with high spatial frequency patterns to estimate SVBRDFs.

In this project, we describe a generic and simple approach to estimate both the topography and SVBRDFs of complex surfaces within the structured light setup.

2 OUR METHOD

Different parametric models have been used to represent SVBRDFs. One commonly used physically-based model was developed by Ward [Ward 1992]. The Ward model characterizes the BRDF of a surface into three components: diffuse reflectance, specular reflectance, and micro-scale roughness, which are typically represented as image-based maps. We use the structured light setup, equipped with polarizers, to estimate the specular and diffuse reflectance components of the SVBRDF of an imaged surface. Then, we use fringe projection profilometry [Huang et al. 2002] to derive height and normal maps. To estimate the roughness component of the SVBRDF, we first observe that for a smooth, homogenous surface, the magnitude and rate of change of specular intensity vary with surface roughness and illumination/view angles. We then observe that a pro-cam system like the one used in our structured-light setup illuminates and views a surface from a regular family of illumination and view angles. Finally, we observe that we can use the projective geometry of our pro-cam system along with the normal maps estimated by the system to “unscramble” the specular maps associated with different materials and thereby estimate the roughness components of their BRDFs.

For a given surface point, the normal at that location governs the outgoing reflection direction. For a fixed view location, the intensity measured at the point represents a sample of the specular lobe. The sample will have maximum intensity for the peak of the lobe if the reflected and view directions coincide. We call the hypothetical normal that would lead to coincidence of view and reflected vector as the *virtual normal*. The estimated location of the sample on the lobe is related to the transformation required to convert the measured surface normal into the virtual normal as is shown in figure 2.

Thus, sorting the measured surface normal at each pixel by the transformation required to match the virtual normal provides a sparse sampling of the 3D isotropic BRDF of the material.

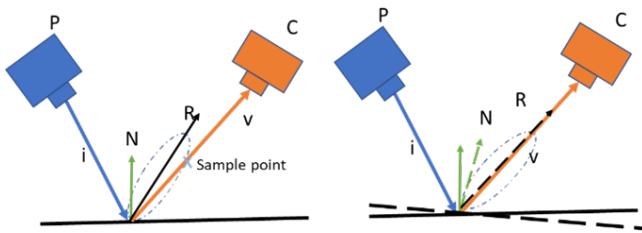


Figure 2: (Left) Every point on a similar material gives a sample of the BRDF due to illumination/viewing geometry. (Right) The sample location is related to the orientation of virtual normal with respect to the measured surface normal.

3 IMPLEMENTATION

First, we calibrate our structured light setup to know the positions and orientations of imaging plane, camera center and projector center in world coordinates. Once, we have estimated normals, we calculate the virtual normals for each pixel using the pro-cam geometry such that the reflection and view vectors align. Next, we use the diffuse map to segment the surface into different materials. We then use this to segment the specular map, and sort the specular intensities in the different segments with respect to the virtual normal. We then plot these sorted specular intensities and fit Gaussian curves to the resulting distributions. The standard deviations of these distributions are proportional to the micro-scale roughnesses of the different materials. To calibrate our estimates to real-world materials, we applied our technique to the NCS gloss standards [NCS 2022] to derive a relationship between measured standard deviations and real-world gloss properties. We then use this relationship to scale our measured standard deviations to surface roughness values. By propagating the roughness values to each pixel in each segment of the surface, we can generate a roughness map of the surface, which along with the height, normal, diffuse and specular maps generated by the system represent a model of complex surface topography and SVBRDF.

4 CONCLUSIONS AND FUTURE WORK

We have presented a technique for estimating the SVBRDFs (including surface roughness) of complex isotropic materials using an off-the-shelf projector-camera system. This technique should be a great addition to the technological toolkit of researchers and practitioners who want to capture and model complex real-world surfaces and use them in physically based rendering. Future work will involve making the approach more robust and applicable to a wide range of materials (especially metals), and adding extensions for translucency and anisotropic reflection.

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